Auteur: Thijs Van Schel
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$$f(x,y) = \sqrt{x^2+y^2}$$

$$a(1,0) \text{ en } h\left(\frac{2\sqrt{5}}{5}, \frac{\sqrt{5}}{5}\right).$$

$$\left\{\begin{array}{l} \dfrac{\partial f}{\partial x}(x,y) = \dfrac{2x}{2\sqrt{x^2+y^2}} = \dfrac{x}{\sqrt{x^2+y^2}} \\ \dfrac{\partial f}{\partial y}(x,y) = \dfrac{2y}{2\sqrt{x^2+y^2}} = \dfrac{y}{\sqrt{x^2+y^2}} \end{array}\right. \Rightarrow \left\{\begin{array}{l} \dfrac{\partial f}{\partial x}(1,0) = \dfrac{1}{\sqrt{1^2+0^2}} = 1 \\ \dfrac{\partial f}{\partial y}(1,0) = \dfrac{0}{\sqrt{1^2+0^2}} = 0 \end{array}\right.$$

$$\Rightarrow Df(a,h) = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{2\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} \end{pmatrix} = \frac{2\sqrt{5}}{5}$$

References